

$$\int_0^1 e^{t^2} dt$$

$$\int_0^x e^{t^2} dt$$

$$\int_0^1 t^2 dt$$

$$\int_0^x t^2 dt$$

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

$$\int_{\pi/4}^{\pi/2} \sin x \, dx$$

$$\int_7^{10} (3e^x + \frac{1}{x}) dx$$

$$\int_0^{\pi/4} \sec^2 \theta d\theta$$

$$\int_0^5 |x^2 - 4x + 3| dx$$

Fundamental theorem of calculus, part 1

Let f be a continuous function defined on $[a, b]$,
and let

$$g(x) = \int_a^x f(t) dt$$

for $a \leq x \leq b$. Then $g'(x) = f(x)$ for all $a < x < b$.

Fundamental Theorem of Calculus, part 2.

Let f be a continuous function defined on $[a, b]$.

Then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f .

Indefinite Integrals and the Net Change Theorem (5.4)

"Indefinite integral" is a traditional name for antiderivative. In this class, we'll use the following definition.

Given a function f defined on an interval, the **indefinite integral** of f is the **general form** of an antiderivative of f on that interval. It is denoted

$$\int f(x) dx$$

$$\int (x^2 + \sqrt{x} + 2\cos x) dx$$

Net change theorem

For any function F which is differentiable on (a, b) ,

$$\int_a^b F'(x) dx = F(b) - F(a)$$

Let $P(t)$ be the population of a bacteria colony at time t .
Suppose

$$\frac{dP}{dt} = Ae^{ct}$$

where A and c are constants. How much does the population increase by between time $t=0$ and $t=10$?

A particle moves along a line with velocity

$$v(t) = t^2 - 4t + 3$$

What is the displacement of the particle between $t = 0$ and $t = 5$?

What is the distance traveled by the particle between $t=0$ and $t=5$?